

Week 47 Gentle step landscapes

Facilitator guide | Final student packets v2 | Prepared 3 October 2026 | Unpiloted

Mathematical overview

Bounded change on a row. Sites are consecutive integers; heights are nonnegative integers, and adjacent heights differ by at most 1. There is at least one fixed clue (c, a_c) . A completion exists exactly when $|a_c - a_d| \leq |c - d|$ for every pair of clues. Distance means the number of steps, not the number of sites visited.

The simultaneous least and greatest completions are

$$L(i) = \max(0, \max_c (a_c - |i - c|)) \text{ and}$$

$$U(i) = \min_c (a_c + |i - c|).$$

Every legal completion lies between them. Both envelopes are themselves legal; thus they also minimize and maximize total height. No fixed supply of blocks or extra height ceiling is assumed. With no clues there is no greatest completion.

Discovery by band. K-1 builds legal landscapes, tests impossible jumps and finds a shortest sequence of one-level moves. Grades 2-3 links allowable heights to distance and optimizes a whole landscape. Grades 4-5 combines bounds and explains why pairwise-consistent clues can always be filled. Experiments support conjectures; the distance and envelope arguments establish them.

Prepare and launch

Use each child's chosen packet, pencils and 5-7 movable markers per board; make clue markers visually different. Flat markers keep zero visible and avoid a large cube supply. Main boards have 22 mm site spacing and 20 mm level spacing; the alternate five-site board uses 25 mm site spacing. Print at actual size. The drawing's top edge is not an extra mathematical restriction.

Readiness. K-1 can compare small whole heights with adult reading. Older entries require counting steps, then adding/subtracting distances; formulas are adult shorthand. With the current ten-child group, keep the three familiar adult-led tables and let readiness determine the page.

Launch, 4 minutes. Let children place markers, then show the printed legal 1,2,2,1 and illegal 1,3,2. Point to the fixed clues and visibly represent height zero. Ask, "Can you change a white marker and keep every neighboring jump small enough?" Do not demonstrate the envelope recipe yet.

Flexible hour. 0-5 launch; 5-22 construct and compare; 22-25 stand and act out a legal up/flat/down walk; 25-45 choose optimization or partner puzzles; 45-55 share one construction and one impossibility reason; 55-60 reset. Completing the packet is not the goal.

Key for the shared starting problems

Problem 1, all bands. Endpoints are 1 at sites 0 and 4. Examples: (1,0,0,0,1), (1,1,1,1,1), (1,2,1,2,1), (1,2,3,2,1). All adjacent differences are at most 1. The middle can be 0 and can be 3. Its full possible range is 0-3, since it is two steps from either clue. Any legal different constructions answer K-1.

Problem 2, all bands. The star is site 2; clues are $h(0)=0$ and $h(4)=2$. Its possible heights are exactly 0, 1, 2. Witnesses are (0,0,0,1,2), (0,0,1,1,2), (0,1,2,2,2). The left clue bounds it above by 2; nonnegativity bounds it below by 0. K-1 asks only for the extrema.

Problem 3. K-1's new right clue is 4: the only landscape is (0,1,2,3,4). In Grades 2-3 and 4-5, 4 is the only right-clue height that forces every site. Larger heights are impossible. For heights $r=0,1,2,3$, there is slack: the least completion and the greatest completion differ, so at least two landscapes exist.

Problem 4, all bands, from top to bottom. First impossible: heights 0 and 3 are only two steps apart. Second possible: (3,2,1,0,0) is a completion. Third impossible: heights 0 and 3 are again only two steps apart. Each impossibility follows by adding two permitted jumps, whose total change cannot exceed 2.

Key for the seven site landscape

Problem 5, all bands. Clues are $h(0)=1$, $h(4)=3$, $h(6)=1$. The least landscape is (1,0,1,2,3,2,1), total 10; the greatest is (1,2,3,4,3,2,1), total 16. Each marker is bounded by its distance from each clue. Both displayed landscapes attain all the respective bounds at once.

For K-1, a third different answer is (1,1,1,2,3,2,1); many answers work. There are exactly 10 completions, independently enumerated, but no printed task asks children to list them all. Grades 2-3 asks for the totals' extrema; Grades 4-5 asks for pointwise extrema. Here the same two constructions solve both questions.

Key for the K-1 finish

Problem 6. The minimum is **5 moves**. A legal route is

(1,2,1,2,1) $\rightarrow$ (1,1,1,2,1) $\rightarrow$ (1,1,1,1,1) $\rightarrow$
 (1,1,0,1,1) $\rightarrow$ (1,0,0,1,1) $\rightarrow$ (1,0,0,0,1).

The three white heights must fall by 2, 1 and 2. One move changes just one height by one level, so fewer than 5 moves cannot suffice. Fixed end clues never move.

Problem 7. Replace, rather than supplement, the old clues. For example, clues $h(0)=0$ and $h(4)=4$ force (0,1,2,3,4); $h(0)=h(4)=1$ allow many answers. Check a child's invented pair by height difference $\leq$ site distance. An impossible puzzle is a valid investigation if it is identified and explained as such.

Key for the older finishes

Grades 2-3 Problem 6. On sites 0-6, $h(0)=0$ and $h(6)=6$ force $(0,1,2,3,4,5,6)$. To gain 6 in six steps, every step must rise by 1. The pair $h(0)=h(6)=0$ allows the all-zero landscape and $(0,1,0,0,0,0,0)$, so it allows more than one. These are examples, not a required classification of all invented pairs.

Grades 2-3 Problem 7. A clue four steps from height 1 can have any whole height **0,1,2,3,4,5**. In general, d steps from a clue of height a , the possible heights are $\max(0,a-d)$ through $a+d$, inclusive, when there are no other clues. Join two allowed clue heights by changing one level until the second height is reached, using flat steps for the remaining distance. Other clues can narrow this range.

Grades 4-5 Problem 6. Sites 0 and 6 are each three steps from the height-2 clue at site 3. At either site every height **0 through 5** is possible. “Possible at a site” is a separate existence claim; arbitrary independent choices at neighboring sites need not fit together.

Grades 4-5 Problem 7. Yes, under the printed nonnegative whole-height rule. Use the lower envelope L from page 1. Each clue contributes a downward tent $a_c - |i-c|$; take the largest of these and zero at every site. The maximum of these 1-Lipschitz functions is 1-Lipschitz. At a clue d , its own tent equals a_d ; pairwise consistency makes every other tent no larger. Thus L has the required clue values and is nonnegative and integral. This is a complete construction, not only a feasibility test.

For completeness, the upper envelope U is also 1-Lipschitz: if every f_c changes by at most one between adjacent sites, their finite minimum does too. Pairwise consistency makes U hit each clue. Every completion h must satisfy $a_c - |i-c| \leq h(i) \leq a_c + |i-c|$, proving the extremal claims.

Optional hints and stopping points

For a stuck construction, ask what one neighbor permits; for impossibility, ask how many steps separate the clues. For an extremum, cover all but one clue and ask what that clue alone forces. For K-1 moves, compare the three starting and target heights before trying to prove optimality. Stop with a convincing construction and a local jump explanation; do not require the envelope notation from children.

Sources and verification

E. J. McShane, *Extension of range of functions*, Bulletin of the AMS 40 (1934), 837-842, [original paper DOI](#). The original PDF was inaccessible during preparation. The extension formula was checked in N. Gigli and E. Pasqualetto, *Equivalence of two different notions of tangent bundle on rectifiable metric measure spaces*, §1.2, equation (1.16), manuscript p. 7: [author manuscript](#). The finite integer proof and keyed examples above were independently checked.

N. Rozhkovskaya, *Math Circles for Elementary School Students*, “Introduction: Berkeley 2009,” describes individual adult support and active parent involvement. The three-table plan here is a local proposal. This guide is unpiloted; digital enumeration and PDF dimension checks are not physical or classroom tests.